

# MOTONGORI HARRIET JOSEPH

Data Scientist | Machine Learning & AI | MSc Data Science Candidate

Nairobi, Kenya | harriet.joseph35@gmail.com

[hattiespartan.github.io](https://hattiespartan.github.io) | [github.com/HattieSpartan](https://github.com/HattieSpartan) | [linkedin.com/in/harriet-joseph-5669b921b](https://linkedin.com/in/harriet-joseph-5669b921b)

## PROFILE

---

Data scientist with an actuarial foundation and 19 months of analytics work inside a live business operation. I use data, statistical modelling and machine learning to understand problems, build solutions and support better decisions. Hands-on with classical ML, NLP and deep learning in Python (scikit-learn, TensorFlow/Keras), from data preparation through evaluation to a working app. MSc Data Science candidate (KCA University) researching statistical learning methods for insurance claims reserving. Building toward machine learning and AI engineering.

## TECHNICAL SKILLS

---

**Machine learning & statistics:** regression, classification (logistic regression, decision trees, random forest, SVM, Naive Bayes), collaborative filtering, cross-validation, model evaluation and error analysis, probability and statistical inference

**Deep learning & NLP:** TensorFlow/Keras (LSTMs, CNNs), text preprocessing and tokenisation, sentiment classification, data augmentation, conversational AI (Rasa)

**Programming & data:** Python (pandas, NumPy, scikit-learn, Matplotlib), SQL, R, data cleaning and validation, exploratory analysis

**Data engineering (developing):** PySpark, Hadoop (HDFS, YARN), Apache Flume, PostgreSQL, Parquet, ETL pipelines

**Analytics & delivery:** Power BI, Tableau, Excel, KPI and root-cause analysis, Streamlit, Git, Zendesk, Snowflake; stakeholder reporting, SOP writing

## SELECTED PROJECTS

---

**Movie-Review Sentiment Analysis with an LSTM** *Python, TensorFlow/Keras, Streamlit* | [GitHub](#)

- Cleaned and tokenised 50,000 IMDB reviews and trained an embedding + LSTM network to classify sentiment, reaching 88% test accuracy.
- Saved the model and tokenizer and wrapped them in a Streamlit app that scores any review typed in, with a confidence value.

**Scene Image Classification with a CNN** *Python, TensorFlow/Keras* | [GitHub](#)

- Built and trained a convolutional neural network to classify ~17,000 images into six scene types, improving test accuracy from 76% to 81% over successive iterations.
- Tested data augmentation and used per-class F1 (0.70–0.95) to trace remaining errors to look-alike classes such as buildings vs streets and glaciers vs mountains.

**Naive Bayes Spam Classifier from First Principles** *Python, probability, scikit-learn* | [GitHub](#)

- Implemented multinomial Naive Bayes by hand, reaching 96.8% test accuracy and catching 93% of spam on unseen messages.
- Benchmarked against scikit-learn (97.9%), traced the gap to how each handles unseen words, and documented the resulting bias.

**Root-Cause Analysis of Operational Performance (CCI Kenya)** *Python (pandas), Matplotlib* | [GitHub](#)

- Analysed ~7,500 performance audits across two sites by week, tenure band, contact reason and scorecard parameter, with agent data anonymised.
- Found a mid-tenure performance dip (60–90-day agents averaged 82.8% vs 89.4% after 90 days) and that six defect types caused 70% of defects; findings set coaching and process priorities.

**End-to-End Big-Data Pipeline (MSc coursework)** *Flume, Hadoop HDFS, PySpark, PostgreSQL*

- Built a pipeline that ingests raw tweets into an HDFS data lake, cleans them in PySpark and loads 219,294 records into a PostgreSQL warehouse over JDBC; in a four-person team, built a second PySpark pipeline writing partitioned Parquet with data-quality rules.

**Also:** customer churn prediction comparing four classifiers ([project](#)); Bella, a Rasa wellness chatbot ([project](#)); a collaborative-filtering movie recommender ([project](#)).

RESEARCH

- MSc thesis (in progress) — KCA University2026 - 2027
- A Comparative Evaluation of Classical Actuarial and Statistical Learning Methods for Claims Reserving in Kenyan Short-Tail General Insurance. Planned design compares chain-ladder methods with machine-learning models under out-of-time validation. Concept paper submitted September 2026.

PROFESSIONAL EXPERIENCE

- Quality Analyst — CCI KenyaSep 2024 - Dec 2025
- Nairobi | Email-based customer operations for a live-shopping marketplace client
- Audited 50–60 customer interactions a day against quality standards, as one of six analysts covering 23–25 agents; tracked QA score (target 85%) and CSAT (target 90%).
  - Used root-cause analysis to find the drivers behind bottom-quartile performance and targeted coaching at them, helping bottom-quartile agents improve by month end.
  - Owned customer-dissatisfaction (DSAT) analysis beyond the role's remit, feeding findings into coaching and process fixes.
  - Took on reporting normally owned by the QA lead: built dashboards and reports used daily and weekly, and presented findings to the campaign's senior manager.
  - Turned recurring defect patterns into SOP revisions and training content; took part in weekly calibration with the client and with team leads.
- Customer Service Representative — CCI KenyaMay 2024 - Sep 2024
- Resolved 60–70 customer emails a day while meeting QA and CSAT targets; promoted to Quality Analyst in under three months.
- Data Analytics Intern — MeriSkill (remote)Sep 2023 - Oct 2023
- Cleaned, explored and visualised datasets in Python and SQL, and presented findings in short written reports.

EDUCATION

- MSc Data Science — KCA UniversitySep 2025 - Nov 2027 (expected)
- Machine learning, NLP, computer vision, big-data engineering.
- Certificate in Data Science — Moringa School2023
- BSc Actuarial Science — South Eastern Kenya University2022
- Statistics, probability, risk and financial mathematics. Treasurer, SEKU Actuarial Students Association (2021–2022).

REFEREES

Available on request.